Transportation & Communication

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Introduction:

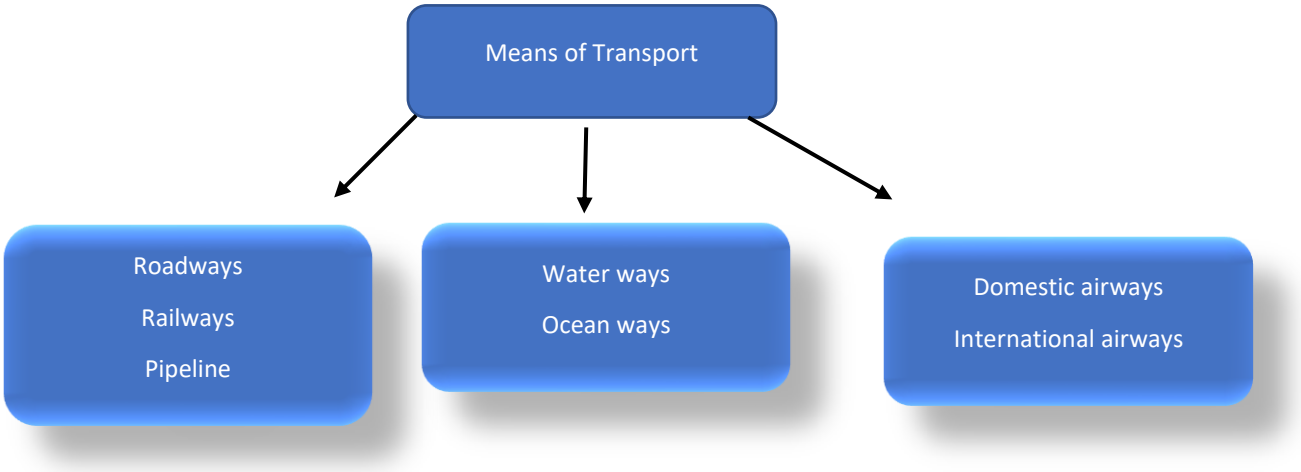
• Transport is indeed a vital component of any country's infrastructure, serving as a lifeline for economic activities and facilitating the movement of people and goods. Here's an overview of the three major means of transport.

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Director,

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Roadways in India

Road Density

- **Densest** – Northern Plains
- **Challenging** – Mountainous regions
- **Highest Density** – Kerala
- **Lowest Density** – Jammu & Kashmir

Classification of Roads

National Highways (NH)
→ Connect major cities/states
→ High-speed, long-distance
→ Central Govt

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State Highways (SH) → Within state, major districts → State Govt
District Roads → Rural ↔ Highways → Local/District authorities
Rural Roads (Village Roads) → Connect villages → Rural development
Border Roads → Border security, strategic use → BRO (Border Roads Organisation)
International Highways → Link with neighboring countries → Trade & travel

Importance of National Highways

- **Backbone** of road transport
- Connect: **Major Cities, State Capitals, Ports, Rail Junctions, Industrial Zones, Tourist Centers**
- Maintained by: **Central Government (NHAI)**

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Shortest National Highway:

- H 548: This highway is 5 km long, making it the shortest National Highway in India.

Andaman and Nicobar Islands:

- NH 4: Known as the Andaman Trunk Road, this highway is located in the Andaman and Nicobar Islands.

East-West National Highway

- NH 27: This highway runs from Porbandar in Gujarat to Silchar in Assam, serving as an important east-west corridor.

States with the Largest Length of National Highways:

- Maharashtra has the largest network of National Highways in India, followed closely by Uttar Pradesh.

Atal Tunnel:

- on October 2020, the Atal Tunnel was inaugurated, known as the world's longest
- highway tunnel at 9.02 km. It connects Manali with Lahaul-Spiti, significantly enhancing accessibility to the region.

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National Highways Development Programme (NHDP)

- **Launched**: 1998
- **Objective**: To develop roadways matching international standards for smooth and efficient traffic flow.
- **Administered by**: National Highways Authority of India (NHAI), created by an Act of Parliament in 1988.
- **Scope**: Covers development, maintenance, and administration of national highways.

Setu Bharatam Programme

- **Launched**: March 4, 2016
- **Budget**: ₹102 billion
- **Goal**: Eliminate all railway crossings on national highways by 2019 to enhance safety and traffic efficiency.
- **Managed by**: Ministry of Road Transport and Highways

Green Highways (Plantation, Transplantation, Beautification & Maintenance) Policy, 2015

- **Aim**:
  - Reduce air pollution and dust.
  - Improve air quality.
  - Prevent soil erosion.
- **Strategy**: Tree and shrub plantation along national highways.

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Golden Quadrilateral

Backbone of road transport

Connect: **Major Cities, State Capitals, Ports, Rail Junctions, Industrial Zones, Tourist Centers**

Maintained by: **Central Government (NHAI)**

Lanes: 4 to 6

Corridor
✓ Delhi-Mumbai
✓ Mumbai-Chennai
✓ Kolkata-Chennai
✓ Delhi-Kolkata

North–South & East–West Corridors

- North–South: **Srinagar → Kanyakumari**
- **East–West: Silchar → Porbandar**
- **Intersection: Jhansi**
- **Objective**: Faster freight & passenger movement

Expressways

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Important Expressways:

- **Mumbai–Pune Expressway**
- **Kolkata–Dumdum Airport Road**
- **Durgapur–Kolkata Road**
- **Yamuna Expressway** (connects **Delhi to Agra** via Greater Noida)

Industrial Corridors

- Chennai-Bengaluru Industrial Corridor
- Bengaluru-Mumbai Economic Corridor
- Vizag-Chennai Industrial Corridor
- Amritsar-Kolkata Industrial Corridor
- Delhi-Mumbai Industrial Corridor

Bharatmala Pariyojana

- **Launched**: Centrally Sponsored
- **Rank**: **2nd largest** after NHDP
- **Target**: **34,800 km** of new roads
- **Phase-I Approval**: **October 2017**
- **Outlay**: ₹5.35 lakh crore
- **Focus**:
  - Economic Corridor Development
  - Border & International Connectivity Road
  - Port Connectivity
  - Greenfield Expressways

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Objectives of Bharatmala

- Improve **efficiency of freight and passenger movemen**
- Optimize existing **road network**
- Enhance **connectivity to backward and tribal areas**
- Reduce **logistics cost**
- Integrate with **Sagarmala**, **industrial corridors**, **logistics parks**

NHAI (National Highways Authority of India)

- **Established**: 1995 (autonomous)
- **Under**: Ministry of Surface Transport
- **Responsible for**: NH maintenance, development, tolling

Indian Railways

- **First railway line** in India was introduced in **1853**, connecting **Bombay (Mumbai) to Thane**.
- **Nationalization** occurred in **1951**, unifying various regional systems under **Indian Railways (IR)**.

➢ Today, Indian Railways is:
The **world's largest** railway network under one management.
The **largest in Asia**.
Known for monumental structures like Chhatrapati Shivaji Terminus and heritage hill railways (e.g., Darjeeling, Nilgiri).

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Railway Zones

Indian railways is geographically organized into zones. At present, there are 18 railway zones

Railway Zones	Headquarters
Northern Railway (NR)	Delhi
North-Western Railway (NWR)	Jaipur
North-Eastern Railway (NER)	Gorakhpur
North-East Frontier Railway (NEFR)	Guwahati
North-Central Railway (NCR)	Allahabad
Southern Railway (SR)	Chennai
South-Western Railway (SWR)	Hubli
South-East Railway (SER)	Kolkata
South-East Central Railway (SECR)	Bilaspur
South-Central Railway (SCR)	Secunderabad
Western Railway (WR)	Mumbai
West-Central Railway (WCR)	Jabalpur
Eastern Railway (ER)	Kolkata
East Coast Railway (ECR)	Bhubaneswar
East-Central Railway (ECR)	Hajipur
Central Railway (CR)	Mumbai
Metro Railway	Kolkata
South Coast Railway	Visakhapatnam

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Classification of Indian Railways:

Broad Gauge

- The distance between rails in broad gauge is 1.676 m, covering 46,807 km and accounting for 74.14% of the total length of rail routes in the country.

Metre Gauge

- The distance between rails in meter gauge is 1 m, with a total length of 13,290 km, constituting 21.02% of the total route length.

Narrow Gauge

- The distance between rails in narrow gauge is 0.762 m or 0.610 m, covering 3,124 km and accounting for 4.94% of the total length of Indian railways.

Metro Rail:

City	Year of Inception
Delhi	2002
Bengaluru	2011
Gurgaon	2013
Jaipur	2015
Chennai	2015
Hyderabad	2015
Kochi	2017
Lucknow	2017
Ahmedabad	2019
Nagpur	2019
Noida	2019
Kanpur	2021

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Mumbai–Ahmedabad Bullet Train Project

- ✓ Project: India's first bullet train route.
- ✓ Initiative: Part of the Make in India programme.
- ✓ Station Launch: Surat station to open by 2023.
- ✓ Operational Start: Expected by 2026

Railway in Kashmir

- ✓ Project: Jammu–Udhampur–Srinagar–Baramulla Railway Link (JUSBRL).
- ✓ Total Length: 345 km (from Jammu to Baramulla).
- ✓ Purpose: Integrates Kashmir with the Indian rail network.

Vande Bharat Express (Train 18)

- **Type:** India's first **indigenously built** semi-high-speed train.
- **Designed by: Integral Coach Factory (ICF), Chennai**.
- **Launched: February 15, 2019**.
- **Make in India Initiative.**
- **Composition:** 16 coaches, 1,128 seating capacity.
- **First Route: Delhi to Varanasi**

Mountain Railways of India (UNESCO & Heritage Sites)

1. **Darjeeling Himalayan Railway** (West Bengal):
   - 2 ft (610 mm) narrow-gauge.

2. **Nilgiri Mountain Railway** (Tamil Nadu):
    - Steam locomotives since **1908**.
3. **Kalka–Shimla Railway** (Himachal Pradesh):
    - 2 ft 6 in (762 mm) gauge; scenic route.
4. **Matheran Hill Railway** (Maharashtra):
    - 20 km from **Neral to Matheran**, Western Ghats.
5. **Kangra Valley Railway** (Himachal Pradesh):
    - 164 km from **Pathankot to Nagar**.

Water Transport in India

- Oldest mode of transport
- Fuel-efficient & eco-friendly
- Ideal for heavy/bulky cargo
- 95% trade by volume
- 70% trade by value
- Top 20 merchant fleets globally
- 12–13 major ports, ~200 minor ports
- Bulk cargo movement
- Coastal shipping
- International & domestic trade
- Cheaper than road/rail
- Environmentally friendly

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Ports in India

- ✓ Located in **9 coastal states**:
  Kerala, Karnataka, Maharashtra, Goa, Gujarat, West Bengal, Odisha, Andhra Pradesh, Tamil Nadu
- ✓ ***Major Ports (13)***
- ✓ **Managed by Central Govt**
- ✓ **India ranks:**
  - **2nd in Asia (Ship-owning)**
  - ✓ **16th globally**

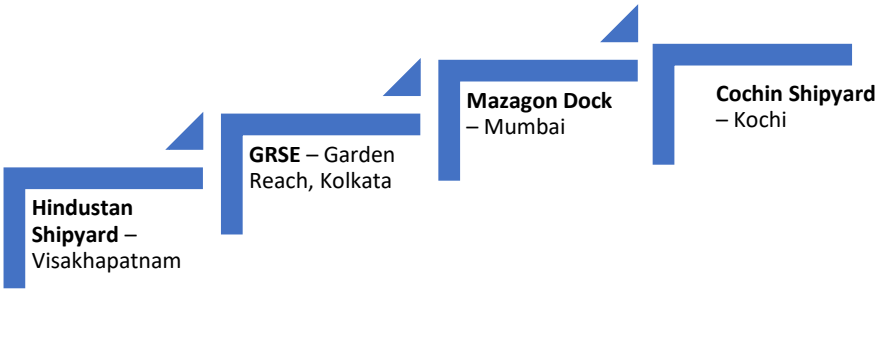


Minor Ports (200+)

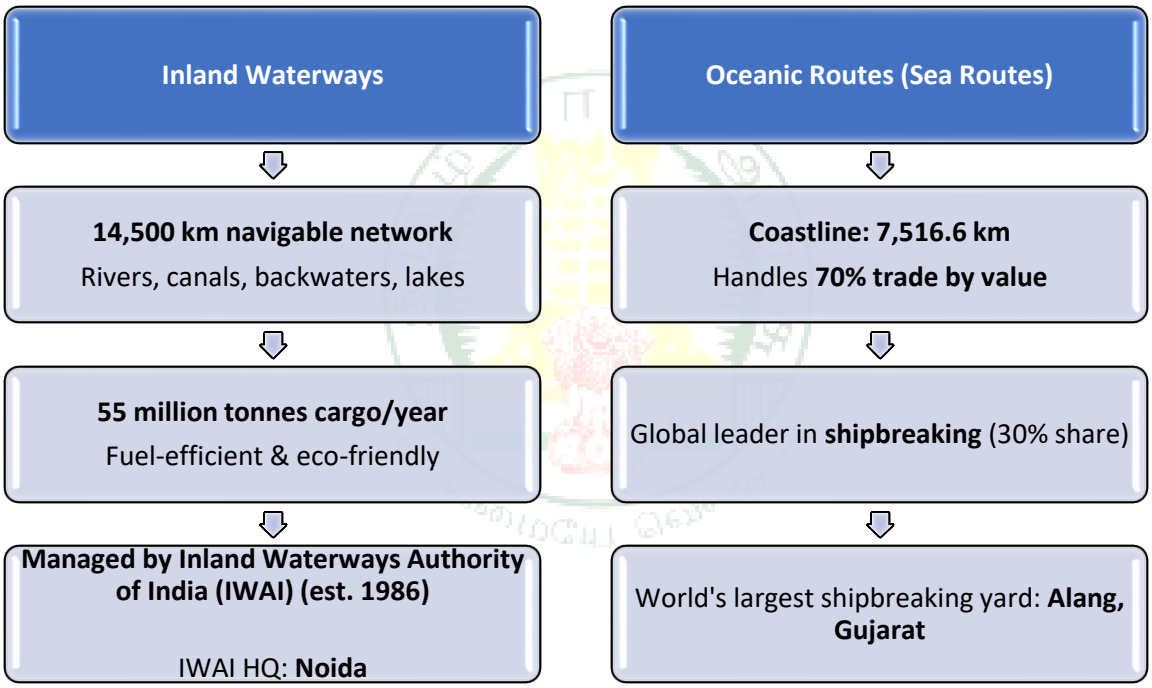
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Managed by State Govts

Shipyards in India



Types of Waterways



Communication

- The postal service
- The postal service was opened to the public in the country in 1837.
- The first Indian postal stamp was issued in 1852 in Karachi.
- Introduced the Quick Mail Service in 1975
- The Quick Mail Service functions on the basis of the system of PIN (Postal Index Number) code which was introduced in 1972.

- Mass Communication Systems
- Electronic Media: Radio broadcasting in India was started in 1923 by the Radio club of Bombay.
- It was named as All India Radio (AIR) in 1936 and again it was renamed as Akashwani in 1957.
- Satellite Communication
- Indian Space Research Organization (ISRO) established in 1969.
- INSAT-1B launched on 30th August 1983 is the first communication satellite in INSAT series.